

Note for Chris on Peristaltic Energy Pumping (or conformal ratchet effect?)

Case 0: Standard hydrodynamics

Consider a locally conserved quantity ρ ,
the hydrodynamic equation of motion
is specified by

$$\partial_t \rho + \vec{\nabla} \cdot \vec{J} = 0$$

continuity

$$\vec{J} = -D \vec{\nabla} \rho$$

Fick's law

$\therefore$ taking D to be constant in space/time
we obtain diffusion

$$\partial_t \rho = D \partial_x^2 \rho$$

Case 2: Particle diffusion with modulated Han

Consider dynamics under the hamiltonian

$$H(t) = f(t) H_0$$

with some conserved particle number $[N, H_0] = 0$

The particle density is governed by

$$\partial_t n + \vec{\nabla} \cdot \vec{J} = 0 \quad \vec{J} = -D \vec{\nabla} n$$

The diffusion constant is modulated in time by the variation in H

$$D = f(t) D_0$$

$\therefore$ in 1D we obtain

$$\partial_t n = D \partial_x^2 n = D_0 f \partial_x^2 n$$

Case 3 Energy diffusion with modulated H

We again consider $H(t) = f(t) H_0$.

The energy density ρ is governed by the

equation

$$\partial_t p = \underbrace{p \partial_t \log f}_{\text{energy generated by modulating } H} + \underbrace{D_0 f \partial_x^2 p}_{\text{diffusion}}$$

to understand the extra term consider the global energy

$$E = \int dx p$$

which evolves as

$$\begin{aligned} \partial_t E &= \int dx \partial_t p \\ &= \int dx p \partial_t \log f + D_0 f \int dx \cancel{\partial_x^2 p} \quad \nearrow = 0 \text{ (total deriv)} \\ &= E \partial_t \log f // \end{aligned}$$

$$\therefore \partial_t \log E = \partial_t \log f$$

$$\therefore E(t) = f(t) E_0 //$$

i.e. the additional term just accounts for the rescaling of energy due to the rescaling of H .

Case 4 : Peristaltic energy pumping

We now consider locally modulated H

$$H = \int dx \underset{\substack{\uparrow \\ \text{local time dependent} \\ \text{terms in } H}}{h(x,t)} = \int dx \underset{\substack{\uparrow \\ \text{spatio-temporal} \\ \text{modulation}}}{f(k(x-vt))} \underset{\substack{\uparrow \\ \text{local time} \\ \text{independent} \\ \text{terms in } H}}{h_0}$$

where $f(z) = f(z + 2\pi)$ (sliding wave profile)



We have energy density

$$\partial_t \rho = \rho \partial_t \log f - \vec{\nabla} \cdot \vec{J},$$

where the current density is

$$\vec{J} = -D \vec{\nabla} \rho = -f D_0 \vec{\nabla} \rho$$

therefore, for 1D we obtain the dynamical equation

$$\partial_t \rho = \rho \partial_x \log f + D_0 \partial_x [f \partial_x \rho]$$

the steady state will be of the form

$$\rho(x, t) = e^{\lambda t} \tilde{\rho}(k(x - vt)), \quad \tilde{\rho}(z) = \tilde{\rho}(z + 2\pi)$$

where λ sets the heating rate

$$E = \int dx \rho(x, t) = e^{\lambda t} \int dx \rho(x, 0) = e^{\lambda t} E_0$$

setting $z = k(x - vt)$ yields a non-dimensionalised equation of motion

$$\tilde{\lambda} \tilde{\rho} - \partial_z \tilde{\rho} = -\tilde{\rho} \partial_z \log f + \tilde{D} \partial_z [f \partial_z \tilde{\rho}]$$

where $\tilde{D} = \frac{D_0 k}{v}$ is the dimensionless diffusion constant.

$\tilde{\lambda} = \frac{\lambda}{k v}$ is the dimensionless heating rate

Perturbative treatment of non-dimensionalised EOM:

The non-dimensionalised equation of motion is a Sturm-Liouville problem, but is still hard to treat exactly.

$\therefore$ We consider the weak driving limit $f = 1 + \Delta f_1$ with $\Delta \ll 1$. Specifically we take

$$\begin{aligned} f &= 1 + \Delta f_1(z) \\ &= 1 + \Delta \sin z \end{aligned}$$

$$\begin{aligned} \tilde{\rho} &= \sum_{n=0}^{\infty} \Delta^n \tilde{\rho}_n \\ \tilde{\lambda} &= \sum_{n=0}^{\infty} \Delta^n \tilde{\lambda}_n \end{aligned}$$

$\therefore$ Expanding the EOM order by order, and extract only the solutions $\tilde{\rho}(z) = \tilde{\rho}(z + 2\pi)$
(the steady state sol'n)

$$\Delta^0: \quad -\partial_z \tilde{\rho}_0 = \tilde{D} \partial_z^2 \tilde{\rho}_0 \quad \therefore \tilde{\rho}_0 = \text{const.}$$

$$\Delta' : \tilde{\lambda}_1 \tilde{\rho}_0 - \partial_z \tilde{\rho}_1 = -\tilde{\rho}_0 \cos z + \tilde{D} \partial_z^2 \tilde{\rho}_1$$

the full solution here is

$$\therefore \tilde{\rho}_1 = z \tilde{\lambda}_1 \tilde{\rho}_0 - c_1 \tilde{D} e^{-z/\tilde{D}} + c_2 - \frac{\tilde{\rho}_0}{1+\tilde{D}^2} (\sin z - \tilde{D} \cos z)$$

as we require $\tilde{\rho}_1(z) = \tilde{\rho}_1(z+2\pi) \therefore \tilde{\lambda}_1 = c_1 = c_2 = 0$

$$\therefore \tilde{\rho}_1 = \frac{\tilde{\rho}_0}{1+\tilde{D}^2} (\sin z - \tilde{D} \cos z)$$

Perturbative current contribution

what is the current contribution?

$$J = -D_0 f \partial_x \rho = -D_0 k f \partial_z \tilde{\rho} = -v \tilde{D} f \partial_z \tilde{\rho}$$

the mean current is obtained by averaging over one period

$$\overline{J} = \frac{1}{T} \int_0^T dt J \quad \text{where} \quad T = \frac{2\pi}{kV}$$

$$= \frac{1}{2\pi} \int_0^{2\pi} dz \, \tilde{J}$$

$$= -\frac{v\tilde{D}}{2\pi} \int_0^{2\pi} dz \, f \, \partial_z \tilde{\rho}$$

$$= -\frac{v\tilde{D}}{2\pi} \int_0^{2\pi} dz \, (1 + \Delta \sin z) (\Delta \partial_z \tilde{\rho}_1 + \Delta^2 \partial_z \tilde{\rho}_2 + \dots)$$

$$= -\frac{v\tilde{D}}{2\pi} \int_0^{2\pi} dz \left[\Delta^2 (\sin z \partial_z \tilde{\rho}_1 + \cancel{\partial_z \tilde{\rho}_2}) + O(\Delta^3) \right] \quad \begin{matrix} = 0 \text{ (total deriv)} \\ \nearrow \end{matrix}$$

$$= -\frac{v\tilde{D}\Delta^2}{2\pi} \int_0^{2\pi} dz \, \sin z \left(\frac{\tilde{\rho}_0}{1+\tilde{D}^2} \right) (\cos z + \tilde{D} \sin z)$$

$$= -\frac{v\tilde{D}\Delta^2}{2\pi} \cdot \frac{\tilde{D} \pi \tilde{\rho}_0}{1+\tilde{D}^2}$$

$\therefore$ the non dimensionalised current $\tilde{J} = \frac{\tilde{J}}{\tilde{\rho}_0 v}$ is given by

$$\tilde{J} = \frac{\tilde{J}}{\tilde{\rho}_0 v} = -\frac{\Delta^2}{2} \cdot \frac{\tilde{D}^2}{1+\tilde{D}^2}$$

recall $\tilde{p}_0 = \bar{p} = \text{mean energy density}$

$\tilde{D} = \frac{D_0 k}{V}$ dimensionless diffusion

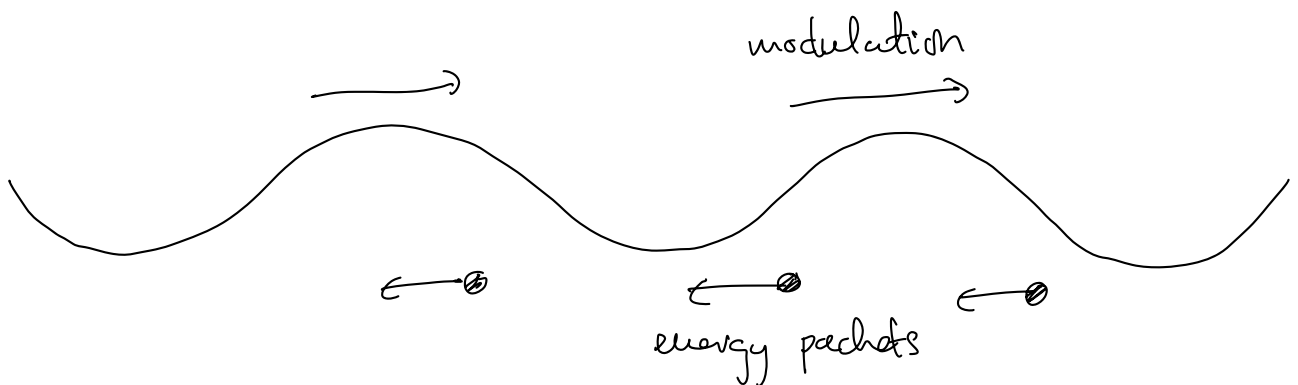
$\tilde{J} = \frac{\bar{J}}{\tilde{p}_0 V}$ dimensionless current

Comments

1) note, perturbative result only valid for

$$\tilde{p}_1 \ll \tilde{p}_0 \quad \therefore \Delta \ll 1, \tilde{D}$$

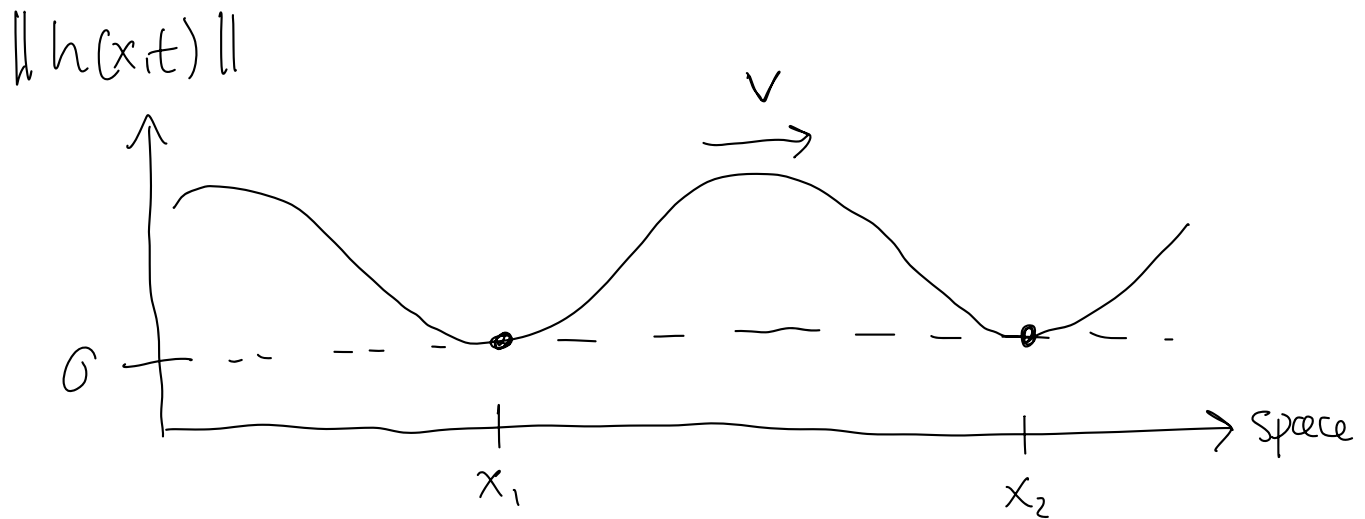
2) interesting: current flows against the sliding wave.



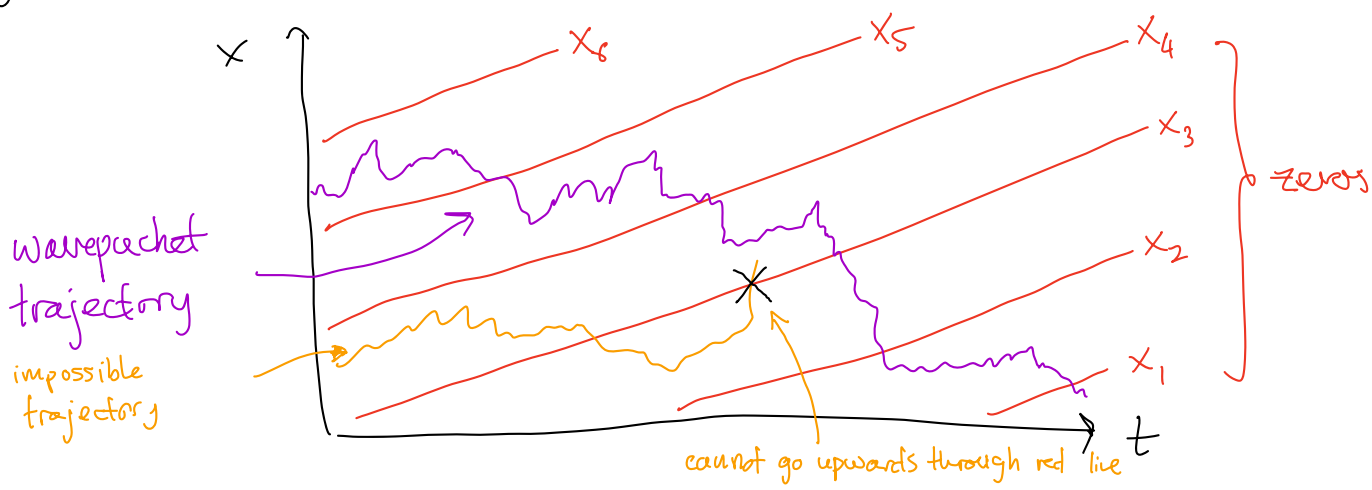
Intuitive picture

Consider the extremal case of $\Delta=1$, where

$f(z)$ has zeros



to energy packets $\in [x_1, x_2]$ the zero at x_2 acts as a reflecting boundary, and the zero at x_1 acts as an absorbing boundary. Specifically: the motion of the profile means a particle at $x = x_{1,2}$ always gets spat out to the left of the zero $x < x_{1,2}$.



$\therefore$ trajectory can pass downwards through zeros,
but not upwards

$\hookrightarrow$ acts as a ratchet

Perturbative heating contribution

To get heating rate we need to solve EOM to $O(\Delta^2)$

$$\Delta^2: \quad \tilde{\chi}_1 \tilde{p}_1 - \partial_z \tilde{p}_2 = (\tilde{p}_0 \sin z - \tilde{p}_1) \cos z + \tilde{D} \partial_z [\sin z \partial_z \tilde{p}_1 + \partial_z^2 \tilde{p}_2]$$

Solving for the solution satisfying $\tilde{p}_2(z) = \tilde{p}_2(z + 2\pi)$ we obtain

$$\tilde{p}_2 = \frac{3\tilde{D}\tilde{p}_0}{4(1+5\tilde{D}^2+4\tilde{D}^4)} \left(3\tilde{D} \cos(2z) + (2\tilde{D}^2-1) \sin(2z) \right)$$

with $\tilde{\chi}_2 = \frac{\tilde{D}}{2(1+\tilde{D}^2)}$

$$\tilde{\chi} = \frac{\tilde{D}\Delta^2}{2(1+\tilde{D}^2)} + O(\Delta^3)$$

Heating rate tied to energy current

$$\tilde{J} = -\tilde{D}\tilde{\lambda} + O(\Delta^3)$$

ie

$$\bar{J} = -\bar{\rho}v \cdot \frac{D_0\lambda}{\sqrt{2}} + O(\Delta^3)$$


natural scale for
current density
($\rightarrow$ current obtained
from sliding all
energy with velocity v)


dimensionless scale
 $\lambda = [\text{time}^{-1}] = \text{heating rate}$
 $= O(\Delta^2)$

$D_0 = \left[\frac{\text{length}^2}{\text{time}} \right] = \text{diffusion const.}$

$v = \left[\frac{\text{length}}{\text{time}} \right] = \text{modulation velocity}$